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Sequence-Oriented Stochastic Model of RO-TRNGs for Entropy Evaluation

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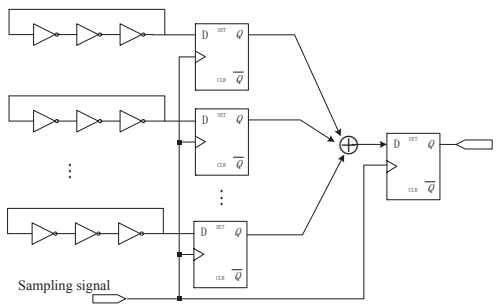
Abstract — Ring oscillator-based true random number generators (RO-TRNGs) are widely used to generate unpredictable random numbers for cryptographic systems. Entropy is usually adopted to quantitatively measure the unpredictability of a TRNG. There have been several stochastic models such as the time-oriented and phase-oriented ones built to evaluate the entropy of Elementary-RO-TRNGs with single oscillator. However, these models are not suitable for the TRNGs composed of multiple oscillators (Multiple-RO-TRNGs), which can obtain more randomness and higher throughput. Considering this, we propose a sequence-oriented stochastic model for the entropy evaluation of RO-TRNGs, named the first-order stationary Markov source model. This model is extensible for the Multiple-RO-TRNGs. Based on that, we present a detailed method to determine the entropy of Multiple-RO-TRNGs. Our proposed model is verified by experiments. Besides, our method can also be a guide to design RO-TRNGs with both high entropy and high throughput.

Key words — True random number generator (TRNG), Ring oscillator, Entropy, Stochastic model, Randomness.

I. Introduction

True random number generators (TRNGs) are important modules in cryptographic systems to provide unpredictable random numbers serving as secret keys, random masks, *etc.*[1,2]. Entropy can be adopt to quantitatively assess the randomness of a TRNG. Ring oscillator-based TRNGs (RO-TRNGs) are a kind of widely used ones on embedded logic devices. Its elementary structure with single oscillator can be shown by Fig.1(a). A fast oscillatory signal is generated from the Ring oscillator (RO) composed of odd number of inverters, then a slow stable clock signal samples it with a D flip-flop. The noises on the logic devices make the oscillatory signal have period jitter[3], resulting in that the sampled bits will be random in $\{0,1\}$, which are taken as the output of the TRNG. For this Elementary-RO-TRNG, designers should do trade-off between the amount of randomness and the throughput. Oscillators can also be constructed in a Multiple RO structure as shown in Fig.1(b), in which, a sampling signal synchronously samples multiple oscillatory signals and all the sampled bits are XORed to gain the final output. This Multiple-RO-TRNG can obtain both large amount of randomness and high throughput[4,5].



(a) An Elementary-RO-TRNG; (b) A Multiple-RO-TRNG

Fig. 1. Structures of RO-TRNGs

In order to evaluate the entropy of RO-TRNGs, several works have proposed stochastic models to mathematically characterize the random behavior of the RO signals. Respectively, Killmann and Schindler proposed a time-oriented stochastic model to evaluate the minimum entropy rate of the RO-TRNGs, which

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modeled the sampling times as a renewal process[6]. Ma *et al.* proposed an equivalent time-oriented model to calculate the entropy rate from specific parameters of the TRNGs[7]. Baudet *et al.* proposed a phase-oriented stochastic model for the entropy evaluation of RO-TRNGs, which modeled the phase of the oscillator signal as a Wiener process[8]. Besides, Amaki *et al.* modeled the state of the sampling points as a Markov chain to calculate the bit patterns probabilities of the output[9], which can be regarded as a state-oriented stochastic model.

By now, the mentioned stochastic models are difficult to extend for the Multiple-RO-TRNGs. Considering that the combination of the oscillators is by XORing the sequences sampled from them, our work proposes a sequence-oriented stochastic model, which is extensible for the Multiple-RO-TRNGs.

In summary, our contributions include:

• We propose a sequence-oriented stochastic model, defined as the first-order stationary Markov source model, to evaluate the entropy of RO-TRNGs.

• Our stochastic model is extensible for the Multiple-RO-TRNGs. Based on this model, we present a method to calculate the entropy rate of Multiple-RO-TRNGs.

• We have verified our stochastic model by experiments. Our entropy evaluation method can also be a guide for the designing of RO-TRNGs with both high entropy rate and high throughput.

II. Preliminaries

We first describe the random behavior of the RO signals under the influence of noises. Then we present some related works on the entropy evaluation of RO-TRNGs.

1. Random behavior of RO signals

In logic devices, the thermal noise is uncorrelated, while the flicker noise is correlated, which is difficult to be characterized by mathematic. For simplicity, most of the stochastic models in previous works have the consistent assumption that only the thermal noise is taken into account and the periods of the oscillatory signals will independently and identically follow Gaussian distribution[3]. Since the two kinds of noise are mutually independent, these models are still valid to estimate the entropy contributed by the thermal noise[10].

The following definition describes the RO signals.

Definition 1 The durations between the flipping times of the oscillatory signal (S_o) are denoted by $X_k, k \geq 1$. The durations between the sampling times of the sampling signal (S_s) are denoted by $T_n, n \geq 1$. The sampled bits sequence are denoted by $B_1, \cdots, B_n, \cdots$. The waiting time $W_n, n \geq 1$ is defined as the duration between the nth sampling point and its following closest edge of S_o.

S_s is stable with $T_n = \mu_s, n \geq 1$. When only the thermal noise is taken into account, all the $X_k, k \geq 1$ of S_o will independently follow an identical Gaussian distribution $N(\mu_o, \sigma_o^2)$. σ_o can be defined as the half period jitter of S_o. Since the jitter is relative between the two signals, for simplicity, we can equivalently regard S_o as a stable signal with $X_k = \mu_o, k \geq 1$ while S_s as an unstable signal with $T_n \sim N(\mu_s, \sigma_s^2), n \geq 1$[7]. In a further step, we normalize all the time variables to μ_o, which is actually the half mean period of S_o and can be easily measured from the frequency of this signal. Then we get the normalized $X_k = 1, k \geq 1$ and $T_n \sim N(\mu, \sigma^2), n \geq 1$. Here $\mu = \frac{\mu_s}{\mu_o}$ and $\sigma = \sqrt{\frac{\mu_s}{\mu_o}} \cdot \frac{\sigma_o}{\mu_o}$, which represents the accumulated jitter in the sampling interval. According to Refs.[6, 7], W_n will approximately follow an uniform distribution in $[0, 1]$, that is $W_n \sim U(0, 1), n \geq 1$. The equivalent random behavior of the RO signals can be presented by Fig.2.

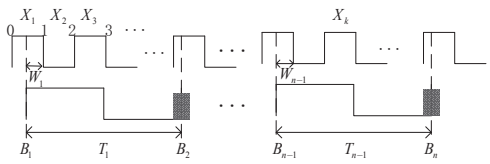


Fig. 2. Equivalent random behavior of the RO signals

2. Related works

For a TRNG that outputs binary sequence $B_1, \cdots, B_n, \cdots$, the n-bit entropy is the Shannon entropy of n-bit output sequence, denoted by $H_n = H(B_1, \cdots, B_n)$. The entropy rate is the average Shannon entropy per bit which can be calculated by

$$H = \lim_{n\to\infty} \frac{H_n}{n} = \lim_{n\to\infty} H(B_n|B_{n-1}, \cdots, B_1) \qquad (1)$$

Related works that aimed to evaluate the entropy of Elementary-RO-TRNGs include:

1) Time-oriented stochastic model. Killmann and Schindler modeled the sampling times as a renewal process[6]. Then they evaluated the minimum entropy rate of RO-TRNGs with the lower bound of $H(B_n|B_{n-1}, \cdots, B_1)$:

$$H_{min} = H(B_n|W) \leq H(B_n|B_{n-1}, \cdots, B_1) \qquad (2)$$

With an equivalent time-oriented model, Ma *et al.* gave a method to calculate H_n and entropy rate $H = \frac{H_n}{n}$, which needs to traverse the 2^n size space of the n-bit sequence[7]. The calculation is massive when n is large.

2) Phase-oriented stochastic model. Baudet *et al.* modeled the phase of the sample point as a Wiener process. They provided a formula to calculate H_n. By

taking $H = \lim_{n\to\infty} \frac{H_n}{n}$, the entropy rate can be calculated with[8]

$$H \approx 1 - \frac{32}{\pi^4 \ln 2} \cos^2(\pi f) e^{-4\pi^2 Q} \tag{3}$$

f is the fractional part of μ and $Q = \frac{\sigma^2}{4}$ is defined as the quality factor of the RO-TRNG.

Ma *et al.* further deduced the H_n to improve its accuracy, then the entropy rate can be calculated by[10]

$$\begin{aligned} H \approx 1 &- \frac{32}{\pi^4 \ln 2} \cos^2(\pi f) e^{-4\pi^2 Q} \\ &- \frac{32}{\pi^4 \ln 2} [1.524 \cos^4(\pi f) - 2.379 \cos^2(\pi f) + 1] e^{-8\pi^2 Q} \end{aligned} \tag{4}$$

3) State-oriented stochastic model. Amaki *et al.* divided one period of the oscillatory signal into several states and adopted Markov chain to characterize the states of the sampling points[9]. This can be regarded as a state-oriented stochastic model. Then they calculated the bit patterns probabilities of the output sequence but had not provide a detailed method to evaluate the entropy.

III. Sequence-Oriented Stochastic Model for Entropy Evaluation

The proposed model is called the first-order stationary Markov source model, which can be used to evaluate the entropy rate of the Elementary-RO-TRNGs.

1. Sequence-oriented stochastic model

We propose to model the Elementary-RO-TRNGs as a first-order stationary Markov source to characterize the random behavior of the output bits sequence, defined by

Definition 2 A first-order stationary Markov source holds the two properties:

a) First-order: for $n \geq 1, b_1 \cdots b_n \in \{0,1\}^n$, $P(B_n = b_n | B_{n-1} = b_{n-1}, \cdots, B_1 = b_1) = P(B_n = b_n | B_{n-1} = b_{n-1})$.

b) Stationary: for $n \geq 1, n' \geq 1, b_{n-1} b_n \in \{0,1\}^2$, $P(B_n = b_n | B_{n-1} = b_{n-1}) = P(B_{n'} = b_n | B_{n'-1} = b_{n-1})$.

First of all, the stationary property of the output sequence of the Elementary-RO-TRNGs has been demonstrated by Killmann and Schindler in Ref.[6]. In order to illustrate the first-order property of RO-TRNGs, we investigate the mutual information of B_n and B_{n-2}, which is denoted by $I(B_n; B_{n-2})$. There is

$$\begin{aligned} I(B_n; B_{n-2}) = &\sum_{b_n, b_{n-2} \in \{0,1\}} P(B_n = b_n, B_{n-2} = b_{n-2}) \\ &\cdot \log_2 \frac{P(B_n = b_n, B_{n-2} = b_{n-2})}{P(B_n = b_n) P(B_{n-2} = b_{n-2})} \end{aligned} \tag{5}$$

in which,

$$\begin{aligned} P(B_n = b_n, B_{n-2} = b_{n-2}) = &\sum_{b_{n-1} \in \{0,1\}} P(B_n = b_n, \\ &B_{n-1} = b_{n-1}, B_{n-2} = b_{n-2}) \end{aligned} \tag{6}$$

Referred to Ref.[8], for the Elementary-RO-TRNG as described in Fig.1(a), there is

$$\begin{aligned} &P(B_n = b_n, \cdots, B_1 = b_1) \\ &= \frac{1}{2^n} + \frac{8}{2^n \pi^2} \Big(\sum_{m=1}^{n-1} (-1)^{b_m + b_{m+1}} \Big) \cos(\pi\mu) e^{-2\pi^2 Q} \\ &\quad + O(e^{-4\pi^2 Q}) \end{aligned} \tag{7}$$

Using the stationary property, we can get $P(B_n = b_n) = \frac{1}{2} + O(e^{-4\pi^2 Q})$, $P(B_{n-2} = b_{n-2}) = \frac{1}{2} + O(e^{-4\pi^2 Q})$, $P(B_n = b_n, B_{n-2} = b_{n-2}) = \frac{1}{4} + O(e^{-4\pi^2 Q})$, then when ignoring the $O(\cdot)$, $I(B_n; B_{n-2}) \approx 0$. Obviously, the mutual information of B_n and $B_{n-2}, \cdots, B_1$ is also approximately equal to 0 , *i.e.*, when B_{n-1} is given, B_n can only depend on B_{n-1}. Consequently, we can model the Elementary-RO-TRNGs as a first-order stationary Markov source.

2. Entropy evaluation

Our method for entropy evaluation is presented by :

Proposition 1 Based on the first-order stationary Markov source model of the Elementary-RO-TRNGs,

a) The transition probabilities of two adjacent bits are

$$\begin{aligned} &P(B_n = 0 | B_{n-1} = 1) = P(B_n = 1 | B_{n-1} = 0) = P_t \\ &= \int_0^1 \sum_{k=-\infty}^{\infty} [C(2k-1+w, f, \sigma) - C(2k+w, f, \sigma)] dw \\ &P(B_n = 0 | B_{n-1} = 0) = P(B_n = 1 | B_{n-1} = 1) v = 1 - P_t \end{aligned} \tag{8}$$

Here $C(\cdot,\cdot,\cdot)$ is the cumulative distribution function (CDF) of Gaussian distribution.

b) The entropy rate can be calculated by :

$$H = -P_t \log_2 P_t - (1 - P_t) \log_2 (1 - P_t) \tag{9}$$

Proof According to random behavior of the RO signals in Section II, there is the conditional probability

$$P(B_n = 0 | B_{n-1} = 1, W_{n-1} = w) = P(1 \leq (T_n - w) \bmod 2 < 2) \tag{10}$$

With $W_{n-1} \sim U(0,1)$, we have

$$\begin{aligned} P_t &= P(B_n = 0 | B_{n-1} = 1) \\ &= \int_0^1 P(1 \leq (T_n - w) \bmod 2 < 2) dw \end{aligned} \tag{11}$$

Since $T_n \sim N(\mu, \sigma^2)$, $\mu = \lfloor \mu \rfloor + f$, f is the fractional part of μ, by defining an auxiliary variable $A = (T_n - \mu) \sim N(f, \sigma^2)$, P_t can be expressed by

$$P_t = \int_0^1 \sum_{k=-\infty}^{\infty} P(2k - 1 + w < A < 2k + w) dw \tag{12}$$

With the CDF of A, we get Eq.(8), so Proposition 1a) is verified. Since $P(A < f - 3\sigma)$ or $P(A > f + 3\sigma)$ is negligible, the k in Eq.(8) can be in a small scale .

In view of the probabilities $P(B_{n-1} = 0)$ and $P(B_{n-1} = 1)$, according to the stationary property of the sequence, we have $P(B_n = 0) = P(B_{n-1} = 0)$ and $P(B_n = 1) = P(B_{n-1} = 1)$. Furthermore, there are

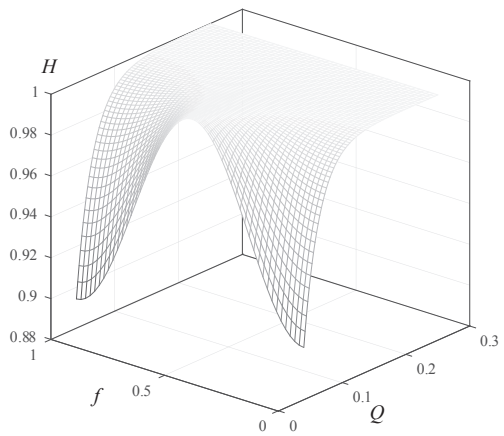
$$\begin{aligned} P(B_n{=}1){=}P(B_{n-1}{=}0)P_t{+}P(B_{n-1}{=}1)(1-P_t) \\ P(B_n{=}0){=}P(B_{n-1}{=}1)P_t{+}P(B_{n-1}{=}0)(1-P_t) \end{aligned} \tag{13}$$

Then we get $P(B_{n-1} = 1) = P(B_{n-1} = 0) = \frac{1}{2}$. Hence, the entropy rate of the TRNG can be calculated by

$$\begin{aligned} H &= \lim_{n\to\infty} H(B_n|B_{n-1},\cdots,B_1) = H(B_n|B_{n-1}) \\ &= \sum_{b_n,b_{n-1}\in\{0,1\}} P(B_{n-1}{=}b_{n-1})P(B_n{=}b_n|B_{n-1}{=}b_{n-1}) \\ &\quad \cdot \log_2 P(B_n{=}b_n|B_{n-1}{=}b_{n-1}) \\ &= -P_t\log_2 P_t - (1-P_t)\log_2(1-P_t) \end{aligned} \tag{14}$$

Then the Proposition 1b) has been verified.

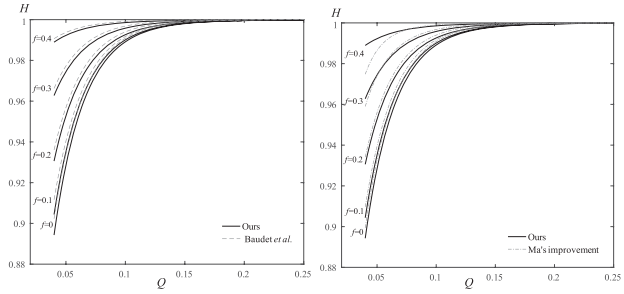
Taking a look into Eqs.(8) and (9) , we can find P_t and H are related with μ's fractional part f and the quality factor $Q = \frac{\sigma^2}{4}$. These relationships are shown by Fig.3. We can see that for all the range of Q, when $f = 0$, P_t is most biased from 0.5 and H reaches to the lowest.



(a) P_t– Q, f (b) H– Q, f

Fig. 3. The relationships of P_t and H with f and Q

In Fig.4, we set $f = 0, 0.1, 0.2, 0.3, 0.4$, then with different Q, we compare the entropy calculated by Eqs.(8) and (9) of our method with the work of Baudet *et al.*[6] presented by Eq.(3) and Ma's improvement[10] presented by Eq.(4). For all the f, our estimation of H is less than that of Baudet *et al.* Besides, considering the worst case that f is close to 0, our estimation result is also less than Ma's improvement. Hence, our method is effective and robust to evaluate the entropy of RO-TRNGs.



(a) Ours vs Baudet *et al.* (b) Ours vs Ma's improvement

Fig. 4. Comparisons of our method with previous ones

IV. Extension for Multiple-RO-TRNGs

First, we describe the Multiple-RO-TRNG by:

Definition 3 For Multiple-RO-TRNG composed of r oscillators denoted by $RO_1\cdots RO_g\cdots RO_r$. The sampled bits sequence of RO_g,$(1 \le g \le r)$ is denoted by $B_{g,1},\cdots,B_{g,n},\cdots$. The output bits sequence of the entire TRNG is denoted by $Y_1\cdots Y_n\cdots$. Then for all $n \ge 1$, there is $Y_n = B_{1,n} \oplus B_{2,n} \oplus \cdots \oplus B_{r,n}$.

The Multiple-RO-TRNG is the combination of several elementary RO sources. To extend our sequence-oriented model to Multiple-RO-TRNG, we assume that every individual RO source has the identical random behavior as described in Section II.1 and can be modeled as a first-order stationary Markov source. In addition, we assume that the component ROs are mutually independent.

Assumption 1 Independence assumption: $RO_1 \cdots RO_g \cdots RO_r$ are mutually independent, *i.e.*, for any $1 \le g \le r$, $1 \le h \le r, h \ne g$, $n, m \ge 1$, $i, j \in \{0, 1\}$, there is $P(B_{g,n} = i|B_{h,m} = j) = P(B_{g,n} = i)$.

By now, our method to calculate the entropy rate of a Multiple-RO-TRNG can be presented by:

Proposition 2 a) The Multiple-RO-TRNG can be modeled as a first-order stationary Markov source.

b) The transition probability is

$$\begin{aligned} P_Y &= P(Y_n = 1|Y_{n-1} = 0) = P(Y_n = 0|Y_{n-1} = 1) \\ &= \sum_{l=0}^{\lfloor (r-1)/2 \rfloor} \binom{r}{2l+1} (P_t)^{2l+1}(1-P_t)^{r-2l-1} \end{aligned} \tag{15}$$

c) The entropy rate can be calculated by

$$H_Y = -P_Y\log_2 P_Y - (1-P_Y)\log_2(1-P_Y) \tag{16}$$

Proof Firstly, according to the first-order property of every individual RO source, when $B_{g,n-1}$ is given, $B_{g,n}(1 \le g \le r)$ can only depend on $B_{g,n-1}$ without $B_{g,m}(1 \le m \le n-2)$. With the additional independence assumption, $B_{g,n}$ will be independent from the $B_{h,m}(1 \le m \le n-2)$ that output by other ROs $(h \ne g)$. That is, when $B_{g,n-1}(1 \le g \le r)$ are given, $B_{g,n}(1 \le g \le r)$ can

only depend on the $B_{g,n-1}(1 \le g \le r)$. Then can we get

$$\begin{aligned} &P(Y_n = Y_n | Y_{n-1} = y_{n-1}, \cdots, Y_1 = y_1) \\ &= P(B_{1,n} \oplus \cdots \oplus B_{r,n} = y_n | B_{1,n-1} \oplus \cdots \oplus B_{r,n-1} = y_{n-1}, \\ &\quad \cdots, B_{1,1} \oplus \cdots \oplus B_{r,1} = y_1) \\ &= P(B_{1,n} \oplus \cdots \oplus B_{r,n} = y_n | B_{1,n-1} \oplus \cdots \oplus B_{r,n-1} = y_{n-1}) \\ &= P(Y_n = y_n | Y_{n-1} = y_{n-1}) \end{aligned} \tag{17}$$

Besides, the transition probability $P(Y_n = y_n | Y_{n-1} = y_{n-1})$ will inherit the stationary property from individual RO. Thus Proposition 2*a*) has been verified. Secondly, in view of the XOR operation in the Multiple-RO-TRNG, when the outputs of odd number of ROs have the transition of $1 \to 0$ or $0 \to 1$, the result of the XOR operation will also have the transition of $1 \to 0$ or $0 \to 1$. Considering that the transition probability of every individual RO is the P_t in Eq.(8), thus we have the transition probability of the entire Multiple-RO-TRNG as

$$P_Y = \sum_{l=0}^{\lfloor (r-1)/2 \rfloor} \binom{r}{2l+1} (P_t)^{2l+1} (1 - P_t)^{r-2l-1} \tag{18}$$

The calculation of H_Y is similar with the Elementary-RO-TRNG. Then *b*) and *c*) of Proposition 2 are verified.

Besides of the parameters f and Q, the entropy rate H_Y of the Multiple-RO-TRNG is related with an additional parameter–r,*i.e.*, the number of the ROs. To illustrate this, considering the worst case that $f = 0$, we vary Q within $0 \to 0.25$ and increase r from 1 to 5, then we calculate the corresponding H_Y. Results in Fig.5 show increasing the number of the ROs will significantly improve the entropy rate. When designing a practical Multiple-RO-TRNG, this result can be the reference for the designers to chose an appropriate number of ROs according to the requirements for entropy rate and throughput.

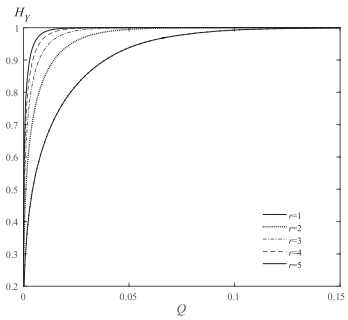


Fig. 5. The relationship of H_Y with r

V. Experiments Verification

We verify our first-order stationary Markov source model by hypothesis testing. For Multiple-RO-TRNGs, we additionally verify the independence assumption.

1. Hypothesis testing

The hypothesis tests respectively verify the first-order Markov and stationary properties of the output sequences. More details are referred to Refs.[11,12].

When we got a set of output sequences from the source, for $(1 \le t \le T)$, value $i, j, k \in \{0, 1\}$, we define $n_i(t)$ is the number of individuals with i at t^{th} bit. $n_{ij}(t)$ is the number of individuals with i at $(t-1)^{th}$ bit, j at t^{th} bit. $n_{ijk}(t)$ is the number of individuals with i at $(t-2)^{th}$ bit, j at $(t-1)^{th}$ bit, k at t^{th} bit. $p_{ij}(t)$ is the transition probability from i at $(t-1)^{th}$ bit to j at t^{th} bit. p_{ij} is the stationary transition probability from i to j. $p_{ijk}(t)$ is the transition probability from i at $(t-2)^{th}$ bit, j at $(t-1)^{th}$ bit to k at t^{th} bit. p_{ijk} is the transition probability from ij to k. Then we have the following hypothesis tests:

a) First-order test. The null hypothesis H_0: $p_{ijk} = p_{jk}$. Estimations for p_{ijk}, p_{jk} are $\hat{p_{ijk}} = \sum_{t=2}^{T} n_{ijk}(t) / \sum_{t=1}^{T-1} n_{ij}(t)$, $\hat{p_{jk}} = \sum_{t=2}^{T} n_{jk}(t) / \sum_{t=1}^{T-1} n_j(t)$. The statistic $-2 \ln \lambda = 2 \sum_{ijk} n_{ijk} (\ln \hat{p_{ijk}} - \ln \hat{p_{jk}}), (n_{ijk} = \sum_{t=2}^{T} n_{ijk}(t))$ has χ^2 distribution with $df = 2 * (2-1)^2 = 2$ degrees of freedom.

b) Stationary test. The null hypothesis H_0 : $p_{ij} = p_{ij}(t)$. Estimations for p_{ij}, $p_{ij}(t)$ are $\hat{p_{ij}} = \sum_{t=2}^{T} n_{ij}(t) / \sum_{t=1}^{T-1} n_i(t)$, $\hat{p_{ij}(t)} = n_{ij}(t)/n_i(t-1)$. The statistic $-2 \ln \lambda = 2 \sum_{t=2}^{T} \sum_{ij} n_{ij}(t) (\ln \hat{p_{ij}(t)} - \ln \hat{p_{ij}})$ will has χ^2 distribution with $df = 2 * (2-1) * (T-1)$ degrees of freedom.

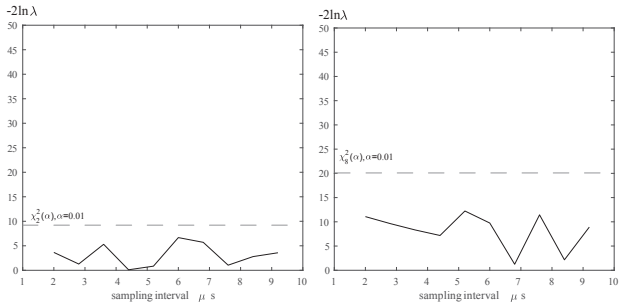
2. Verification for Elementary-RO-TRNGs

We implement a 3-inverters oscillator on an Altera Cyclone IV FPGA, which has about 305MHz frequency. The oscillatory signal is sampled with the intervals from 2μs to 9.2μs with step of 0.8μs. In each case, we set $T = 5$ and acquire a data set of $M * T = 10^6 * 5$ bits. When setting $\alpha = 0.01$, the statistics calculated from the data sets are shown in Fig.6. For the first-order test, there is $-2 \ln \lambda < \chi_2^2(\alpha)$. For the stationary test, there is $df = 8$ and $-2 \ln \lambda < \chi_8^2(\alpha)$. The output bits passed the tests. Therefore, modeling the Elementary-RO-TRNG as a first-order stationary Markov source is acceptable.

3. Verification for Multiple-RO-TRNGs

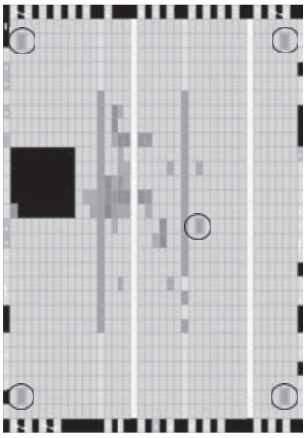
1) Independence verification. As Yoo *et al.* have investigated that when the ROs are far located on the chip, correlations of the ROs could be unnoticeable[13]. Here we design a 5-RO TRNG on the Cyclone IV FPGA. Then we evaluate the correlation coefficient of the outputs of every two ROs in two cases: *a*) all the ROs are locked in the same cell (marked by circle). *b*) the ROs are locked in different cells as far as possible. The layouts

are shown in Fig.7. The results are given in Table 1, in which $CC_{u,v}$ represents the correlation coefficient of RO_u and RO_v. In case a), the $CC_{1,3}$ is much higher than the other correlation coefficients, indicating that the ROs may have high correlation when located near each other. While in case b), the correlation of every two ROs keeps in low level. Hence, it is possible to make the ROs mutually independent by careful layout. The independence assumption for the component ROs is reasonable and practical.



(a) First-order test (b) Stationary test

Fig. 6. Model verification of Elementary-RO-TRNG on FPGA



(a) Case a) (b) Case b)

Fig. 7. Layouts of the two cases

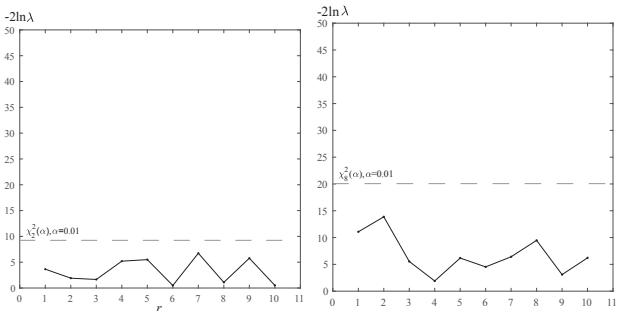
2) Model verification. We also apply the hypothesis tests to verify the first-order and stationary properties of the Multiple-RO-TRNGs. We implement the TRNGs composed of different number of ROs ($r : 1 \to 10$) on the FPGA. The sampling interval is set as 2μs, we acquire and test the outputs of every TRNG instance. The testing results in Fig.8 show that the output sequences passed the tests, hence the Multiple-RO-TRNGs can also be modeled as first-order stationary Markov sources.

VI. Application for RO-TRNG Designing

We present a Multiple-RO-TRNG prototype to show the application of our entropy evaluation method on TRNG designing.

Table 1. Correlation coefficients

Correlation coefficients	case a) (near)	case b) (far)
$CC_{1,2}$	0.0180	-0.0147
$CC_{1,3}$	**0.8643**	0.0025
$CC_{1,4}$	0.0003	0.0069
$CC_{1,5}$	0.0015	0.0043
$CC_{2,3}$	0.0172	0.0021
$CC_{2,4}$	0.0015	0.0070
$CC_{2,5}$	0.0080	0.0187
$CC_{3,4}$	0.0001	0.0113
$CC_{3,5}$	0.0019	0.0017
$CC_{4,5}$	0.0045	0.0016



(a) First-order test (b) Stationary test

Fig. 8. Model verification of Multiple-RO-TRNG on FPGA

The implementation platform is the same Cyclone IV FPGA device. This prototype aims to gain 5Mbit/s throughput high entropy random bits. Firstly, we adopt the method proposed by Haddad *et al.*[14] to measure the accumulated jitter σ of a 3-inverters oscillator with the sampling frequency of 5MHz. The measurement result of σ is about 0.1249 and we can calculate $Q = 0.0039$. Then with our entropy evaluation method, we figure out that the number of the ROs should be more than 13 to ensure the entropy rate is larger than 0.997/bit (required by AIS31[15]). Besides, in order to make the component ROs approximately independent with each other, we locate the ROs in the cells as the layout in Fig.9. This 13-RO TRNG prototype implemented on the FPGA has at least 0.997/bit entropy rate and a 5Mbits/s throughput. In addition, we acquire 24000 output bits of this TRNG and do statistical tests. The output successfully passed the NIST SP800-22 tests[16]. Hence, this prototype also has good statistical properties and does not even need post-processing for optimizing its output. In practical use, we recommend to adopt on-line tests to detect intolerable weakness of the TRNG following the AIS31 and NIST SP800-90B[17] documents.

VII. Conclusions

In this paper, we work on the entropy evaluation of RO-TRNGs. We propose a sequence-oriented stochastic model called the first-order stationary Markov source model, which can be extended to Multiple-RO-TRNGs.

Results of our experiments demonstrate that this model is effective for RO-TRNGs. Based on this model, we present the method to calculate the entropy rate of a Multiple-RO-TRNG from the jitter parameter and the number of the component ROs. In addition, our method can help to design a Multiple-RO-TRNG according to the requirements for entropy rate and throughput.

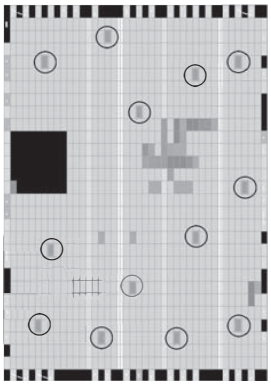


Fig. 9. Layout of the Multiple-RO-TRNG prototype on FPGA

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